
Data Analysis Project

Data Analyst:

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Client/Sponsor:

Banking/fintech financial institution with SME portfolio

Purpose:

This project aims to investigate credit risk in SBA 7(a) loans (US, 2020–2025) in order to improve the origination decision. We will identify higher risk segments by industry (NAICS), geography (State), loan size, and term, and build a simple and explainable logistics model (PD) that sorts applicants by likelihood of default.

Based on that ranking (deciles), we will define an action threshold (cut-off) to prioritize revision/limit or term adjustments and estimate the expected impact on charge-off and expected loss ($EL \approx PD \times LGD_proxy \times EAD$). The result will be communicated with cohort/vintage KPIs, maps/tables by segment, and an executive memo with actionable recommendations for Risk, Origination, and Management.

Scope / Main activities of the project:

Activity	Description
Data collection	Collect data from the SBA 7(a) FOIA 2020–present dataset
Sort by Probability of Default	Sort requests by risk with a simple PD
Identify deciles with the highest defaults	Analyze the deciles that concentrate the defaults and set a threshold that activates review/adjustments
Create dashboard with risk–volume trade-off	Present clear dashboard showing the trade-off between risk and volume
Deliver final report	Delivery of the final report with recommendations for the

	financial institution.
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This project does not include:

- Any area outside the boundaries of the United States.
- Implementations of any solution or recommendation.
- Loan data of more than 5 years in the project (2020–2025)

Deliveries:

Deliverable	Description/ Details
Scope of Work	A clear statement of the business task you've selected to investigate
Sources collected	A description of all data sources used
Data cleaning or handling documentation	Documentation of any data cleaning or manipulation
Documentation of findings	A summary of your analysis
Dashboard and key findings	Supporting visualizations and key findings
Extra documentation	Based on what you discover, a list of additional deliverables that you find useful to include for further exploration
Final Report	Your top high-level insights based on your analysis

Schedule Summary / Main Milestones:

Milestone	Expected completion date	Description/Details
<i>Data Review</i>	<i>24/08/25</i>	<i>Review of data sources completed</i>
<i>Data analysis</i>	<i>24/08/25</i>	<i>Initial data analysis completed</i>
<i>List of deciles</i>	<i>25/08/25</i>	<i>Deciles with the highest default rate identified</i>
<i>Threshold Creation</i>	<i>25/08/25</i>	<i>Review/adjustment threshold set</i>
<i>Data Visualization</i>	<i>25/08/25</i>	<i>Dashboard with trade-off and top KPIs completed</i>
<i>List of recommendations</i>	<i>26/08/25</i>	<i>List of recommendations for each decile</i>
<i>Final Report</i>	<i>27/08/25</i>	<i>Final report detailing all the work done, analyses, methodologies, and discoveries.</i>

*Estimated completion date:

August 27, 2025